

The Portfolio Scenario:
National Healthcare Resource Allocation – Phase 3
(Data Analysis Project)

Data Analyst: Symone Thompson
Client/Sponsor: Public Health Infrastructure & Logistics Analytics
Target Asset: 03_descending_high_volume_hotspots.csv

Executive Summary & Objective:
The objective of Phase 3 is to isolate and rank the primary maternal healthcare delivery bottlenecks across the United States. By forcing a descending sort order over the complete historical CDC natality dataset, this analysis extracts the highest-priority "hotspots" experiencing dense volumes of live births. This information equips decision-makers with the operational intelligence needed to adjust medical funding allocations, deploy specialized medical staff, and optimize hospital bed capacities where infrastructure strain is greatest.

Purpose:
This phase represents the core analytical engine of the project portfolio, executing targeted queries to shift focus away from sparse distributions and directly isolate major urban and regional clinical centers. The findings provide immediate visibility into year-over-year operational fluctuations inside dominant healthcare markets (such as Los Angeles County, CA, Harris County, TX, and Cook County, IL).

Scope / Major Project Activities:
The major parts of this project. High-level steps, activities, or stages of the project, and a brief description for each.

Activity	Description
Environment Setup & Data Ingestion	Access the Google Cloud Platform sandbox, initialize a BigQuery workspace console, and connect to the public

	<code>sdoh_cdc_wonder_natality</code> data ecosystem. Ingestion of historical CDC Wonder Natality county logs via the Google Cloud Platform sandbox environment.
Schema Assessment & Data Hygiene	Application of explicit descending sort constraints (<code>`ORDER BY Births DESC`</code>) over millions of public health logs.
SQL Transformation & Aggregation	Isolation and structural extraction of the top 10 high-density national delivery markets.
Insight Generation & Business Reporting	Chronological tracking of year-over-year volume changes within identified hotspot counties between 2016 and 2018.

This project does not include:

- **Total County Populations:** This analysis strictly measures raw birth volumes (`Births`) rather than calculating per capita birth rates against the total population of each county.
- **Gender Distinctions:** The scope isolates total infant delivery counts without filtering or segregating data based on the sex assigned at birth.
- **Stillborn and Mortality Metrics:** The dataset exclusively records live natality events; fetal deaths, stillbirths, and infant mortality tracking are completely excluded from this scope.
- Tracking of low-density regional data frameworks (fully executed during Phase 2 operations).
- Direct deployment of live API data pipelines to production hospital networks.

Deliverables:

A specific list of things that this project will deliver.

Column Name	Data Type	Analytical Purpose
<code>Year</code>	DATE	<i>Tracks the chronological year to monitor growth or decline in high-priority sectors.</i>
<code>County_of_Residence</code>	STRING	<i>Identifies the primary localized hotspot county requiring immediate infrastructure review.</i>
<code>Births</code>	INTEGER	<i>The core metric used to calculate volume strain and prioritize logistical hospital resources..</i>

Ave_Age_of_Mother	FLOAT	Supplies clinical context regarding regional parental demographic profiles.
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Schedule Overview / Major Milestones:

Project execution commences on June 18th, 2026, structured across highly focused technical milestones.

Milestone	Expected Completion Date	Description/Details
Milestone 1:	June 18th, 2026	Query Drafting & Ascending Constraint Validation — Completed within 24 Hours.
Milestone 2:	June 19th, 2026	System Extraction to Google Sheets & CSV Formatting — Completed within 48 Hours.
Milestone 3:	June 21st, 2026	Deployment of Dataset 02 to Local Project Directories — Completed within 72 Hours.

*Estimated date for completion:

Final delivery, asset deployment, and repository verification are scheduled for **June 22nd, 2026**.

The Analytical Imperative:

In healthcare logistics, data is directly tied to human outcomes. This analysis transcends simple numbers on a screen; by uncovering a clear, year-over-year baseline in high-volume sectors like Los Angeles County, it provides decision-makers with the proactive runway required to shift medical funding, optimize hospital bed capacities, and ensure that maternal care resources are precisely deployed before local medical infrastructure faces critical bottlenecks. True data maturity turns historical logs into protective, life-saving strategies.